CAVENDISH UNIVERSITY UGANDA

Campus Compass

Smart Room Booking & Campus Navigation System

SYSTEM MANUAL & PITCH DOCUMENT | IEEE CUU Student Branch | 2026

"Know the room. Own the day."

01 — THE PROBLEM

What We Are Solving

Every day at Cavendish University Uganda, students and lecturers waste valuable time because there is no reliable way to find and book a room.

Lecture Room Clashes — Lecturers arrive to find their assigned room already occupied. Timetable clashes are only discovered when two groups show up at the same door — causing disruption and wasted contact hours.

The Revision Room Struggle — Students organising group study and revision sessions — especially during exam season — have no way to check which rooms are vacant or reserve a space in advance. They wander the campus losing study time they cannot afford to lose.

Manual, Slow Coordination — Class representatives must physically visit the admin office or make phone calls just to request a room. Notice boards and spreadsheets are updated too slowly to reflect real availability.

Empty Rooms, No Visibility — Rooms sit empty because nobody knows they are available. The university has the physical space — the problem is that this information is invisible to the people who need it most.

02 — THE SOLUTION

How Campus Compass Works

Campus Compass is a Progressive Web App (PWA) — no download, no installation. Any student or staff member opens it in their browser and immediately has access to live room availability, bookings, schedules, and notifications.

Find Any Space — Live — Search lecture halls, seminar rooms, labs, and discussion rooms by building, capacity, or equipment. Availability is shown in real time — no guessing, no walking around.

- **Instant Revision Room Booking** — Students browse all available spaces, filter by group size and facilities, and reserve a revision or study room in seconds — from their phone, without visiting any office. The system automatically prevents double bookings and sends a confirmation.
- **Clash-Free Lecture Scheduling** — Priority rules, capacity checks, and database-level exclusion constraints guarantee that two sessions can never occupy the same room at the same time. Timetable conflicts are caught automatically before they cause disruption.
- **QR Check-In at the Door** — Every room has a door display screen. Students and lecturers scan a rotating QR code on arrival to confirm attendance — secure, instant, and privacy-friendly.
- **Physical, Online & Hybrid Sessions** — The system supports all three modes. Online meeting links and passcodes are delivered securely only to authorised participants — never on public displays or notice boards.
- **Smart Notifications** — Booking confirmations, reminders, cancellations, and no-show alerts are delivered via email, SMS (Africa's Talking), and browser push notifications.
- **Room Door Display Screen** — Each room's display shows the current session, the next booking, live occupancy status, and a QR code for check-in. Any screen (TV, tablet, monitor) can be a door display.
- **Admin Control Dashboard** — Department administrators manage rooms, timetables, users, bookings, and notifications from a single dashboard. Super admins have full oversight across all departments.

03 — HOW TO USE IT

Step-by-Step User Journey

Student Booking a Revision Room

Open Campus Compass — Visit the app URL on any phone or laptop — no installation needed.

Log in with Student ID — Enter your university ID and password. A one-time PIN is sent for verification.

Browse Available Rooms — Tap "Find a room", filter by date, time, capacity, and room type.

Book in Seconds — Select a room, enter session name and group size, confirm. Clash check is instant.

Receive Confirmation — A confirmation is sent via email and SMS with all booking details.

Check In on Arrival — Scan the QR code on the room's door display to confirm attendance.

Lecturer Confirming a Lecture Room

Log in — Sign in with your staff ID. Upcoming lectures appear on the home screen immediately.

View Today's Schedule — See all assigned rooms, times, and lecture details — before leaving home.

Get Reminders — Automatic reminders are sent 30 minutes before each session via SMS and push notification.

Arrive and Check In — Scan the door QR code. The room display updates to show your lecture as active.

Share Online Link — For hybrid/online sessions, a secure meeting link is sent only to enrolled students.

04 — USER ROLES

Who Uses Campus Compass

👨‍🎓	**Student**	Browse rooms, book revision spaces, view timetable, join lectures, check in at the door.
📣	**Class Representative**	Book rooms on behalf of their class, manage class bookings, coordinate revision sessions.
👩‍🏫	**Lecturer**	View assigned lecture rooms, check schedule, manage sessions, check in on arrival.
🏢	**Department Admin**	Manage rooms, approve bookings, create timetable slots, oversee department activity.
🔑	**Super Admin**	Full control over all departments, users, facilities, settings, and operational reports.
🖥️	**Door Display**	A dedicated screen mode for room entrances — current session, next booking, live QR code.

05 — FEATURE OVERVIEW

What the System Includes

Feature	Who Benefits	Status
Real-time room availability	Students, Lecturers, Admins	✓ **Live**
Revision room booking	Students, Class Reps	✓ **Live**
Lecture room scheduling	Lecturers, Dept Admins	✓ **Live**
QR code check-in	Students, Lecturers	✓ **Live**
Door display screen	All campus visitors	✓ **Live**
Email & SMS notifications	All users	✓ **Live**
Hybrid / Online session support	Lecturers, Students	✓ **Live**
No-show auto-release	All users	✓ **Live**
Admin dashboard	Dept Admins, Super Admin	✓ **Live**
Offline support (PWA)	All users	✓ **Live**

06 — THE IMPACT

What This Means for Cavendish University

29	**335**	**0**
Rooms across campus already in the	Modules mapped to rooms and	App downloads needed — works in any

system	timetable slots	browser

⏱ Time Saved — Students and lecturers no longer spend time walking campus looking for rooms or queuing at administration offices. Every booking takes under 60 seconds.

🏫 Better Space Utilisation — Rooms are released automatically when no one checks in, ensuring available spaces are visible and bookable — not locked away in a spreadsheet.

🏘 Fairer Access for Students — Every student has equal, transparent access to book a revision room — not just those who know which administrator to call.

⚡ Reduced Admin Workload — Reminders, no-show releases, clash detection, and booking confirmations all run automatically. Administrators focus on decisions, not manual work.

07 — TECHNOLOGY

Built for Africa, Designed to Scale

Campus Compass is built with proven, production-grade technology and deployed on free-tier cloud services — meaning it costs very little to run and can scale to the entire university without significant IT investment.

Node.js 20 **Express API** **PostgreSQL** **Neon (Cloud DB)** **Vercel (Frontend)** **Render (Backend)** **Progressive Web App (PWA)** **Africa's Talking SMS** **Web Push Notifications** **QR Code Check-In** **GitHub CI/CD** **Africa/Kampala Timezone**

Why a Progressive Web App (PWA)?
A PWA works on any phone or laptop browser — Android, iPhone, Windows, or Chromebook. No Play Store, no App Store, no storage space needed. Students on limited data plans benefit from intelligent caching that makes the app work even with poor connectivity. This removes every barrier to adoption at Cavendish University Uganda.

What Makes Campus Compass Worth Exploring

☑ It Already Works

Campus Compass is not a concept or a wireframe. It is a fully functional system with 29 rooms, 335 modules, 31 lecturers, and 21 programmes already loaded — built specifically for Cavendish University Uganda.

Solves a Real Daily Problem

Every student and lecturer on campus experiences the room problem every single day. This is not a theoretical pain point — it is a real, measurable cost to teaching quality and student productivity.

Low Cost to Deploy

The entire system runs on free-tier cloud services. The university does not need to buy servers, install software, or hire a dedicated IT team to run it.

Scales With the University

As Cavendish grows — more students, more rooms, more departments — Campus Compass scales with it. Labs, seminar rooms, and any bookable facility can be added easily.

Built for Uganda

Africa's Talking SMS integration, Kampala timezone, and offline PWA support make this system designed for the real conditions students and staff face on campus — not imported from a different context.